ML Model Evaluation Metrics – Basics

Model evaluation metrics help us determine **how well a machine learning model performs, especially for classification problems**:

The most basic concepts are:

Confusion Matrix

- Accuracy
- Precision
- Recall
- F1 Score
- FNR
- FPR
- Specificity
-

1. **Confusion Matrix**

- A confusion matrix **compares the actual class with the predicted class**.
- For a binary classification problem such as Pass/Fail:

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

1. **Confusion Matrix Meaning:**

(i) **TP – True Positive**

- Actual = Positive, Predicted = Positive.
- Example: A student actually passes and the model predicts Pass.

(ii) TN – True Negative

- Actual = Negative, Predicted = Negative.
- Example: A student actually fails and the model predicts Fail.

(iii) FP – False Positive

Actual = Negative, Predicted = Positive.

Example: Student actually fails, but model predicts Pass.

This is also called a Type I error.

(**iv) FN – False Negative**

- Actual = Positive, Predicted = Negative.
- Example: Student actually **passes**, but model predicts **Fail**.
- This is also called a **Type II error**.

2. **Accuracy**

Accuracy tells us:

What proportion of all predictions were correct?

Formula: Accuracy=(TP+TN)/(TP+FP+FN+TN)

Example

Suppose: TP = 40 , TN = 45 , FP = 5 ,FN = 10

Then: Accuracy=?

When is accuracy useful?

3. **Precision**

- Precision: It measures the exact number of true positive cases divided by the total number of positive predictions made.
- Precision=TP/(FP+TP)

Precision answers:

Of all patients predicted as Diseased, how many were actually diseased?

4. **Recall**

- Recall is also called Sensitivity or True Positive Rate.
- Recall=TP/(FN+TP)
- It answers: Of all the patients who actually have the disease, how many were correctly identified by the model?

5. **F1 SCORE**

- F1 score combines Precision and Recall into a single metric.
- It is the **harmonic mean of precision and recall.**

$F1=2\times Precision x Recall/(Precision+Recall)$

Using: Precision=0.8889 Recall=0.8

Therefore: $F1=2\times(0.8889x0.80)/(0.8889+0.8)$ $F1\approx0.8421$

General interpretation of F1-score

F1-score	General interpretation
90–100%	Excellent
80–89%	Very good
70–79%	Good / acceptable in many applications
60–69%	Moderate
Below 60%	Generally needs improvement

Simple memory trick

Accuracy → Overall correctness

Precision → Predicted Positive correctness

Recall → Actual Positive coverage

F1 → Balance between Precision and Recall

Confusion Matrix → TP, TN, FP, FN from which the metrics are calculated

6. **FPR (False Positive Rate)**

- **FPR (False Positive Rate)** measures the proportion of **actual negative cases that the model incorrectly predicts as positive**.
- For example FPR measures the proportion of healthy patients incorrectly classified as diseased.
- FPR=FP/(TN+FP)

7. **FNR(False Negative Rate)**

- **FNR (False Negative Rate)** measures the proportion of **actual positive cases that the model incorrectly predicts as negative**.
- For example FNR measures the proportion of diseased patients incorrectly classified as healthy.
- FNR=FN+(TP+FN)

8. **Specificity**

- Specificity measures how well a classification model correctly identifies the negative cases.
- Specificity=TN/(TN+FP)

A machine learning model was developed to diagnose a particular disease based on patient data. The model was tested, and the following confusion matrix was obtained:

Actual \ Predicted	**Diseased (D)**	**Healthy (H)**
Diseased (D)	40	15
Healthy (H)	10	135

a. Calculate the accuracy, precision, recall, and F1-score.

b. Determine the false positive rate (FPR) and false negative rate (FNR).

Suggest ways to improve the model's performance based on the confusion matrix

Calculations

- For your **disease diagnosis example**, the F1-score is: F1=76.19%
- This can be considered **reasonably good**, but it should **not automatically be considered adequate for medical diagnosis**.
- The important issue is that:

  Recall=72.73%

  which means the model is missing about **27.27% of diseased patients**.

For medical diagnosis

- Recall/sensitivity is often more important than F1-score because false negatives can be serious. Therefore, I would evaluate:
- Recall/Sensitivity → How many diseased patients are detected?
- F1-score → Balance between precision and recall
- FPR → How many healthy patients are incorrectly diagnosed?

➢ An F1-score of 76.19% indicates reasonably good performance, but for a disease-diagnosis application, further improvement is desirable, particularly in recall, because the current FNR of 27.27% means that a considerable number of diseased patients are being missed.